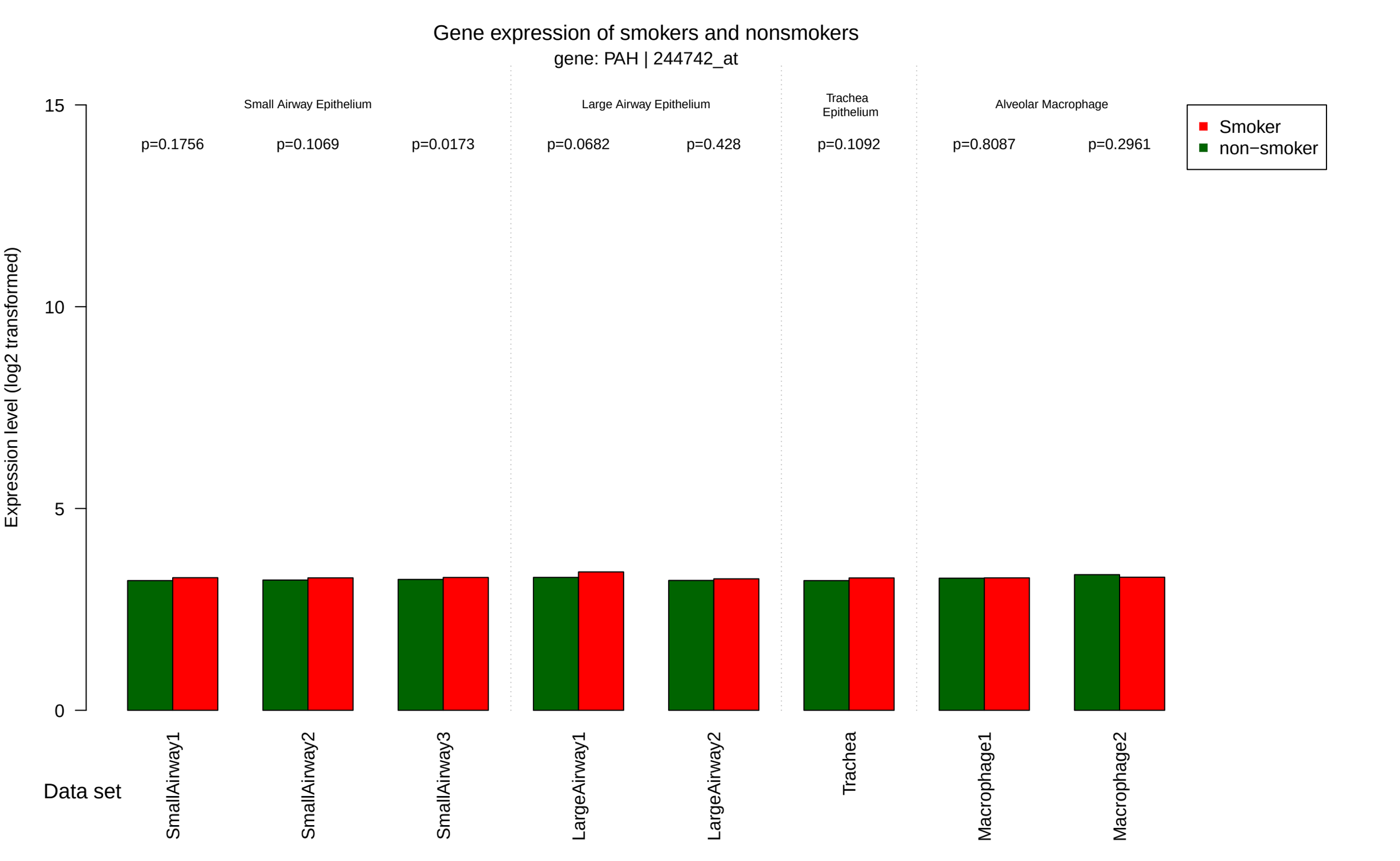
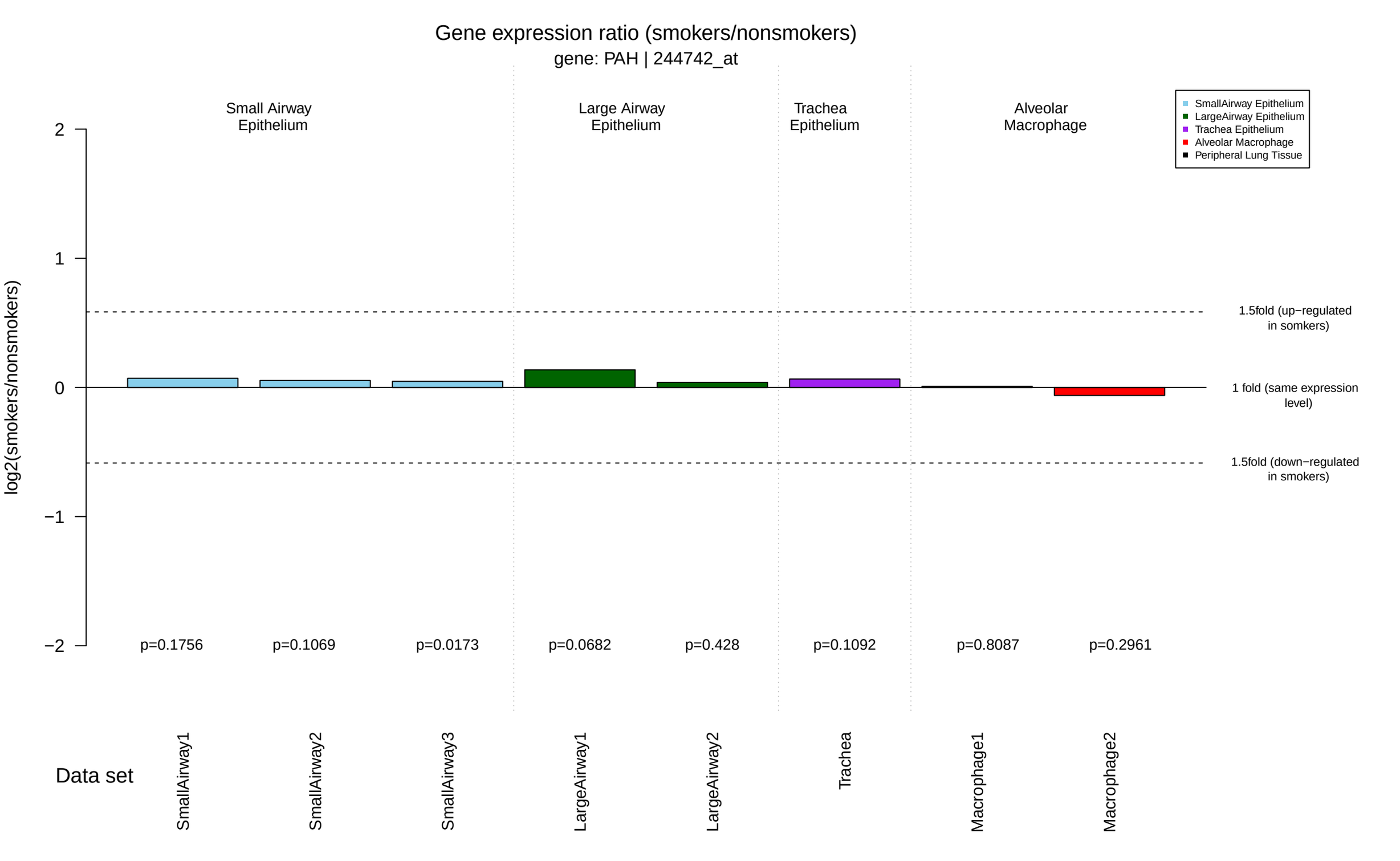
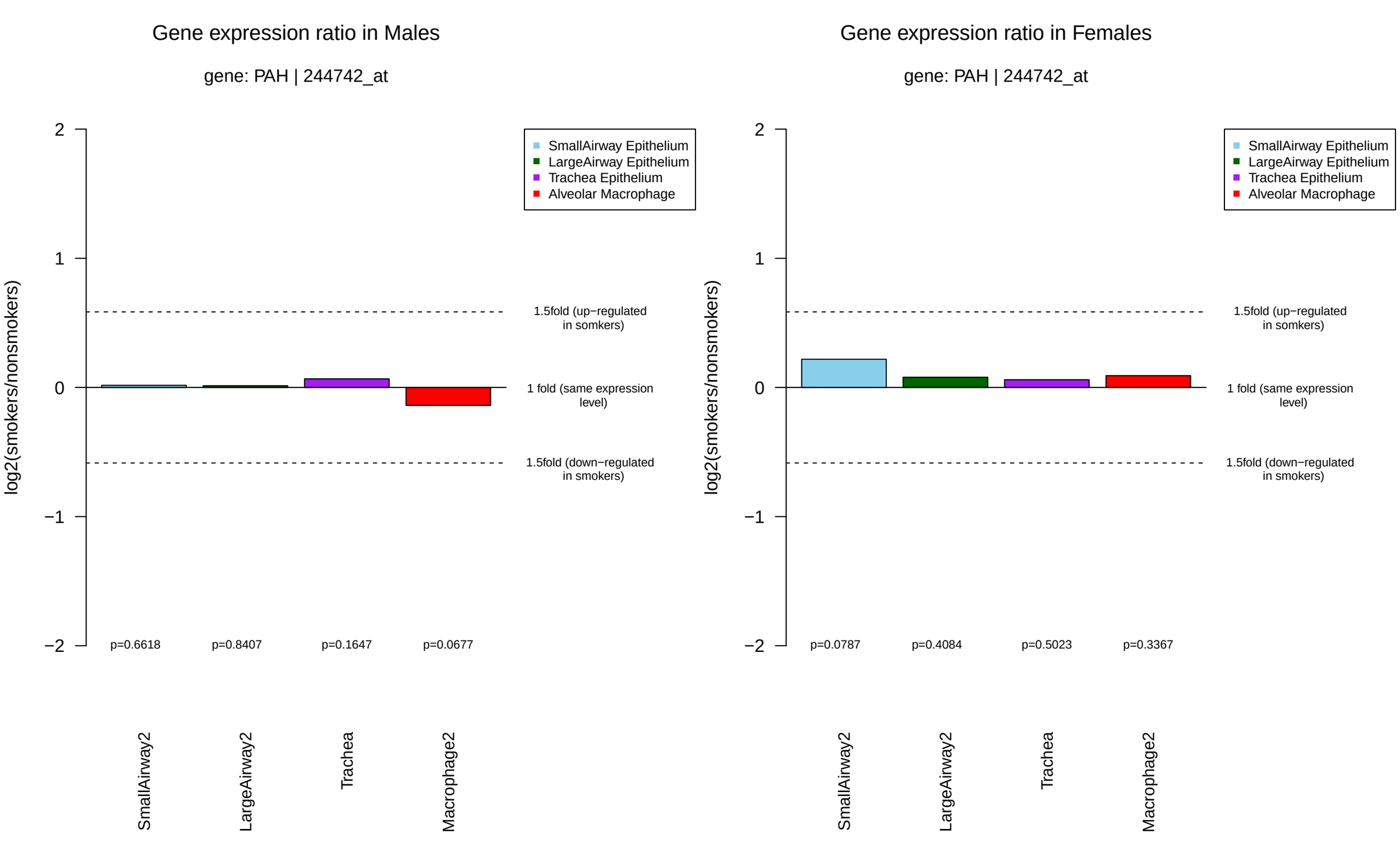
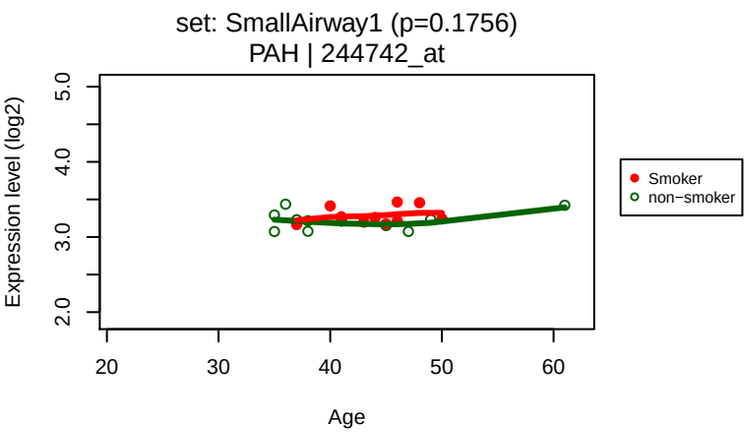
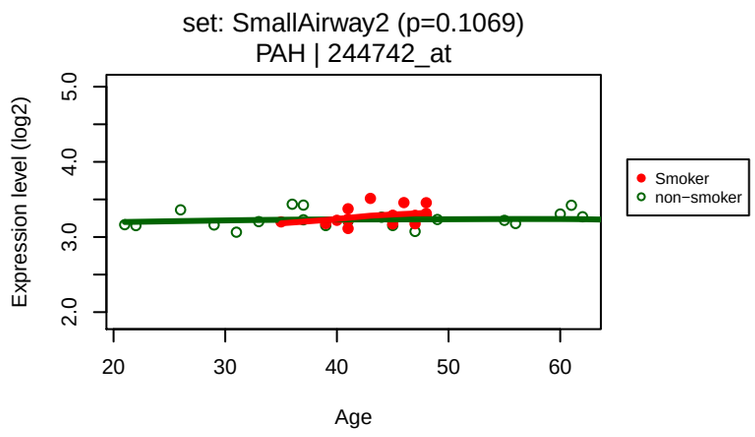
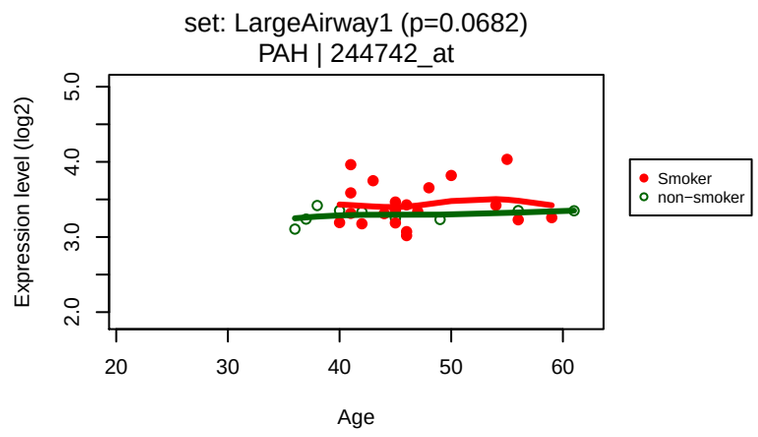

Gene expression of smokers and nonsmokers
gene: PAH | 244742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1756
p=0.1069
p=0.0173
p=0.0682
p=0.428
p=0.1092
p=0.8087
p=0.2961
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

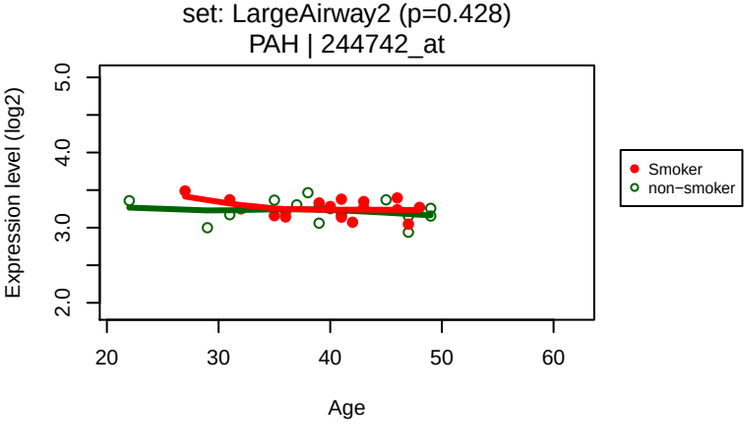

Gene expression ratio (smokers/nonsmokers)
gene: PAH | 244742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1756
p=0.1069
p=0.0173
p=0.0682
p=0.428
p=0.1092
p=0.8087
p=0.2961
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

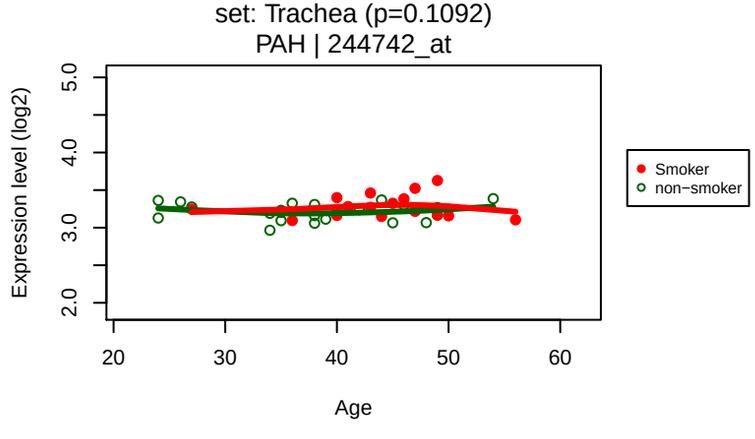

Gene expression ratio in Males
gene: PAH | 244742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6618
p=0.8407
p=0.1647
p=0.0677
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAH | 244742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0787
p=0.4084
p=0.5023
p=0.3367
SmallAirway2
LargeAirway2
Trachea
Macrophage2

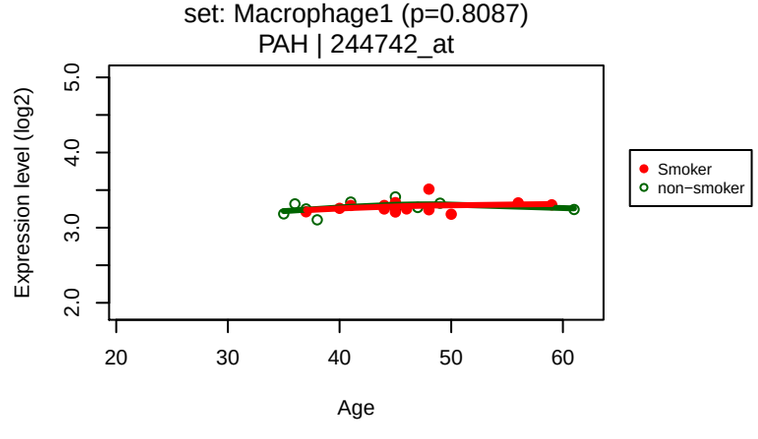

set: SmallAirway1 (p=0.1756)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker

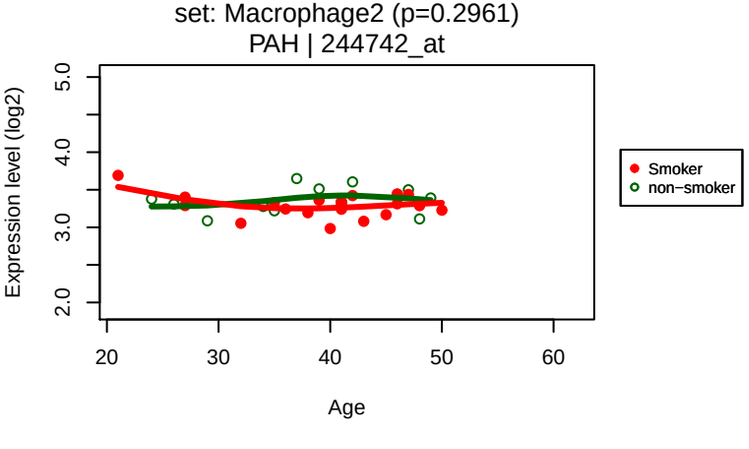

set: SmallAirway2 (p=0.1069)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker

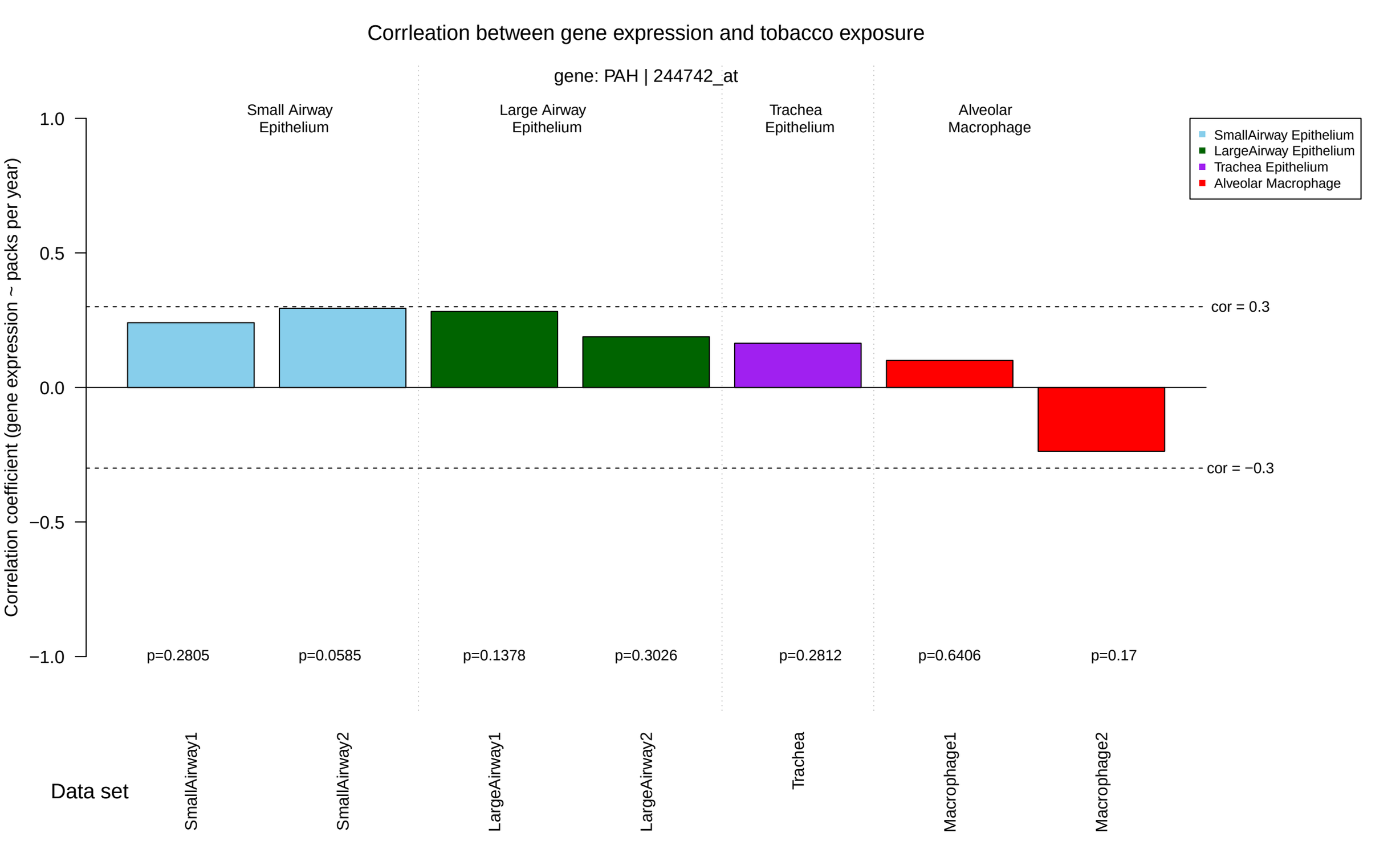

set: LargeAirway1 (p=0.0682)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.428)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1092)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8087)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2961)
PAH | 244742_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAH | 244742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2805
p=0.0585
p=0.1378
p=0.3026
p=0.2812
p=0.6406
p=0.17
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2